mochammad wildan 202343501453 R6R prediksi saham buy or sell dengan metode klasifikasi dan decision tree

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

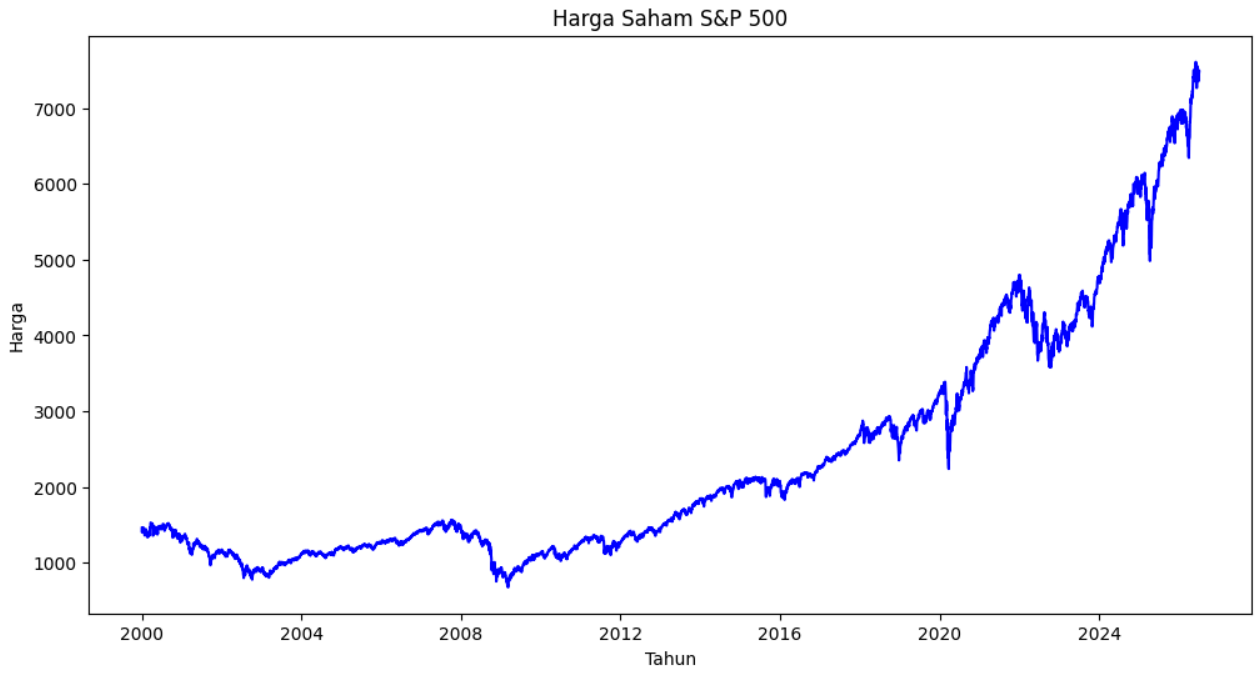
```
df = pd.read_csv('/content/sample_data/global_financial_markets_2000_Now.csv')
print("dataset berhasil dimuat!")
```

```
dataset berhasil dimuat!
```

```
#'^GSPC' (S&P 500)
df_saham =df[df['symbol'] == '^GSPC'].copy()
```

```
df_saham['date'] = pd.to_datetime(df_saham['date'])
df_saham = df_saham.sort_values('date')
```

```
plt.figure(figsize=(12, 6))
plt.plot(df_saham['date'], df_saham['close'], color = 'blue')
plt.title('Harga Saham S&P 500')
plt.xlabel('Tahun')
plt.ylabel('Harga')
plt.show()
```




```
df_saham['target'] = (df_saham['close'].shift(-1) > df_saham['close']).astype(int)
df_saham = df_saham.dropna()
print(f"total data siap di gunakan : {len(df_saham)} baris.")
```

```
total data siap di gunakan : 6664 baris.
```

```
from sklearn.model_selection import train_test_split

features = ['open', 'high', 'low', 'volume']
X = df_saham[features]
y = df_saham['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
from sklearn.tree import DecisionTreeClassifier


model = DecisionTreeClassifier(max_depth=5, random_state=42)
model.fit(X_train, y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(max_depth=5, random_state=42)
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

y_pred = model.predict(X_test)
print(f"Akurasi model: {accuracy_score(y_test, y_pred)*100:2f}%\n")
print(classification_report(y_test, y_pred))
```

```
Akurasi model: 52.813203%

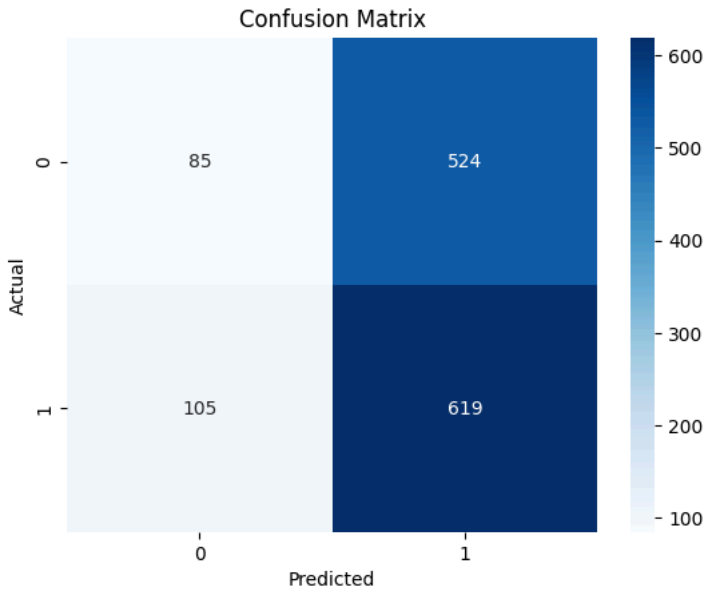
              precision    recall  f1-score   support

           0       0.45      0.14      0.21       609
           1       0.54      0.85      0.66       724

    accuracy                           0.53      1333
   macro avg       0.49      0.50      0.44      1333
weighted avg       0.50      0.53      0.46      1333
```

```
from sklearn.tree import plot_tree

cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```




```
plt.figure(figsize=(20,10))
plot_tree(model, feature_names=features, class_names=['sell','buy'], filled=True)
plt.show()
```

